Modeling Thermal Stresses in 3-D IC Interwafer Interconnects

Jing Zhang, Max O. Bloomfield, Jian-Qiang Lu, Ronald J. Gutmann, and Timothy S. Cale

***Abstract*—We present a finite-element-based analysis to determine if there are potential reliability concerns due to thermally induced stresses in interwafer copper via structures in three-dimensional (3-D) ICs when benzocyclobutene (BCB) is used as the dielectric adhesive to bond wafers. We first partially validate our approach by comparing computed results against two types of experimental data from planar ICs: 1) volume-averaged thermal stresses measured by X-ray diffraction in an array of parallel Cu lines passivated with TEOS and 2) studies of failures induced by thermal cycling via chain structures embedded in SiLK or SiCOH. In the volume-averaged thermal stress study, predicted stress slopes $(d\sigma/dT)$ agree well with other modeling results. Our computed stress slopes agree reasonably well with experimental data along the Cu line direction and normal to the Cu lines surface, but we underestimate the stress slope across the Cu line. In the case of via chains, computed von Mises stresses agree with the results of thermal cycle experiments; we predict failure when SiLK is used as a dielectric and predict no failure when SiCOH is used as the dielectric. The approach is then employed to study thermal stresses in interwafer Cu vias in 3-D IC structures bonded with BCB. Simulations show that the von Mises stresses in interwafer Cu vias decrease with decreasing pitch at constant via size, increase with decreasing via size at constant pitch, and decrease with decreasing BCB thickness. We conclude that there is a concern regarding the stability of interwafer Cu vias. Guidelines for design parameter values are estimated, e.g., interwafer via size, pitch, and BCB thickness. For 2.6-μm-thick BCB, computations indicate that via size should be larger than 3 μm at a pitch of 10 μm to avoid plastic yield of Cu vias.**

***Index Terms*—Benzocyclobutene (BCB), Cu, interwafer vias, low-k dielectronics, thermo-mechanical stresses, three-dimensional (3-D) ICs.**

I. Introduction

WAFER-LEVEL three-dimensional (3-D) integration offers improved performance and functionality over conventional planar integrated circuits (ICs). A primary driver for 3-D ICs is the promise of reduced signal delay through shortened interconnects [6]. Moreover, monolithic wafer-level 3-D integration promises increased functionality through the integration of diverse technologies, while maintaining the cost advantage of monolithically fabricated interconnects [8]. Fig. 1 shows a schematic of an approach to 3-D integration developed at Rensselaer Polytechnic Institute (RPI) [10]. A processed wafer (top) is bonded to another processed wafer (bottom) using benzocyclobutene (BCB). The top wafer is backside thinned to a few microns of Si. Cu interwafer interconnects are formed by interconnecting specified points in the multilevel metallization (MLM) layers of these two wafers by Cu damascene patterning [10].

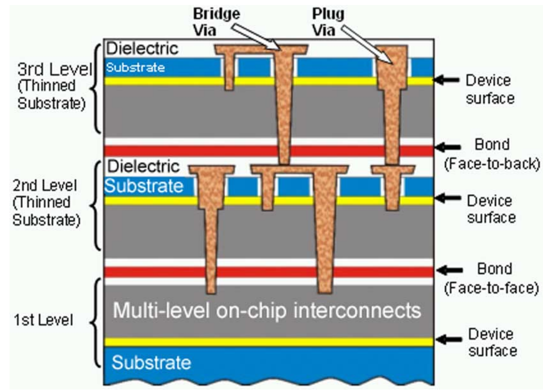

Fig. 1. 3-D chip cross-sectional structure using "face-to-face" bonding [3].

Among the design and/or processing concerns surrounding 3-D ICs is the mechanical stability of the structures [11]. It is not currently feasible to reliably predict total stresses in ICs, or adhesion of the layers in ICs; it is even difficult to relate structural failures to specific stress levels. Tests have been done to show that BCB bonding and thinning, as done in the RPI process flow, does not impact (planar) IC performance [12], and the wafers do not delaminate during standard reliability testing [13]. It is clear that the BCB-bonded wafers are stable; however, the wafers in those studies did not have interwafer vias. One open question is the effect of thermally induced stresses and whether they are significant during processing and during 3-D IC operations on interwafer Cu vias, as they can be for planar ICs [3], [5]. Thermally induced stresses can be predicted using structural models that represent proposed structures reasonably well and the coefficients of thermal expansion (CTEs) of the materials present. In this paper, we use finite-element (FE)-based modeling to address the question of thermally induced stresses in interwafer Cu vias to determine if they are a cause for concern for the stability of 3-D ICs and whether more quantitative work is warranted. Computations using our FE model clearly indicated that there is reason to be concerned about the reliability of interwafer Cu

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J. Zhang was with the Focus Center—New York, Rensselaer: Interconnections for Gigascale Integration, Rensselaer Polytechnic Institute, Troy, NY 12180 USA. He is now with the Department of Mechanical Engineering, University of Alaska, Fairbanks, AK 99775-5960 USA.

M. O. Bloomfield, J.-Q. Lu, R. J. Gutmann, and T. S. Cale are with the Focus Center—New York, Rensselaer: Interconnections for Gigascale Integration, Rensselaer Polytechnic Institute, Troy, NY 12180 USA (e-mail: calet@rpi.edu).

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vias. Since there is no experimental information from 3-D ICs against which to compare predicted results, we validated our approach using studies on planar ICs.

Thermal stresses in MLM structures in planar ICs have been studied both experimentally and by modeling. In X-ray diffraction (XRD)-based studies [14]–[16], stresses in the metal lines can be derived from the measured strains as determined by measuring changes in lattice spacing. Wafer curvature methods [17], [18] measure the changes in the curvature of the substrate on which a blanket film is deposited. These two experimental approaches are useful for comparing with averages of stresses predicted by models. On the other hand, they are not useful for identifying localized failures, e.g., individual failed vias. Direct imaging of the failures in carefully prepared samples using scanning electron microscopy (SEM) provides direct evidence for "qualitative comparison" with model predictions. By qualitative we mean that failures are determined but not the stresses at which those failures occur. FE-based analyses can provide predictions of detailed stress distributions and have been used to model the stresses and deformation in interconnect structures. Some models are based on the two-dimensional (2-D) plane strain assumption [19]–[22], in which vias and/or metal lines extend infinitely in and out of the plane. Three-dimensional models [21], [23], [24] have been also used to simulate thermal stress distributions in interconnect structures with complicated geometries. If predicted stresses exceed, or even approach, the yield strength of one or more materials, then there is cause for concern about the stability. Of course, it should be kept in mind that several assumptions routinely used in such analyses make quantitative comparison with experiment tenuous [21], [23], and thermally induced stresses are only part of the total stress.

This paper is organized as follows. We first partially validate our modeling approach by comparing computed results against data from two types of experiments: 1) volume-averaged thermal stresses measured by XRD in an array of parallel Cu lines passivated with tetraethyl orthosilicate (TEOS) sourced oxide [2] and 2) stacked vias structures with SiLK [3] or carbon-doped silicon oxide (SiCOH) [5] as the dielectric. Then, we present modeling results from our study of thermal stresses in interwafer Cu vias that would form part of 3-D ICs, as in the process developed at RPI; i.e., bonded with BCB. Guidelines for design parameter values are estimated, e.g., interwafer via size and pitch and the thickness of BCB (see Fig. 1).

II. Experimental Information

In this section, we review results from two types of experiments from the literature that we use to at least partially validate our approach to studying thermal stresses: 1) volume-averaged thermal stresses in arrays of parallel Cu lines measured by XRD and 2) a stacked via chain structure.

As described in [2], volume-averaged thermal stresses of single damascene Cu lines passivated with TEOS oxide were determined from strains measured using XRD. The test structures are arrays of parallel Cu lines fabricated using damascene processing. Four sets of lines are reported on, with nominal widths/pitches: 1.0/2.0, 0.6/1.2, 0.4/0.7, and 0.25/0.55 μm. SEM cross sections are not provided in [2] and [19], so we

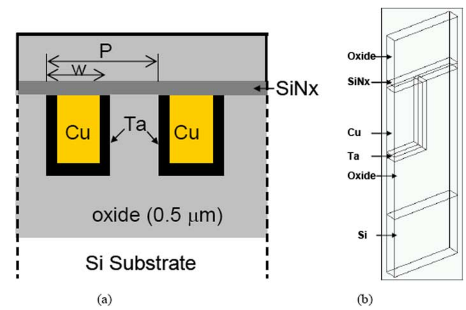
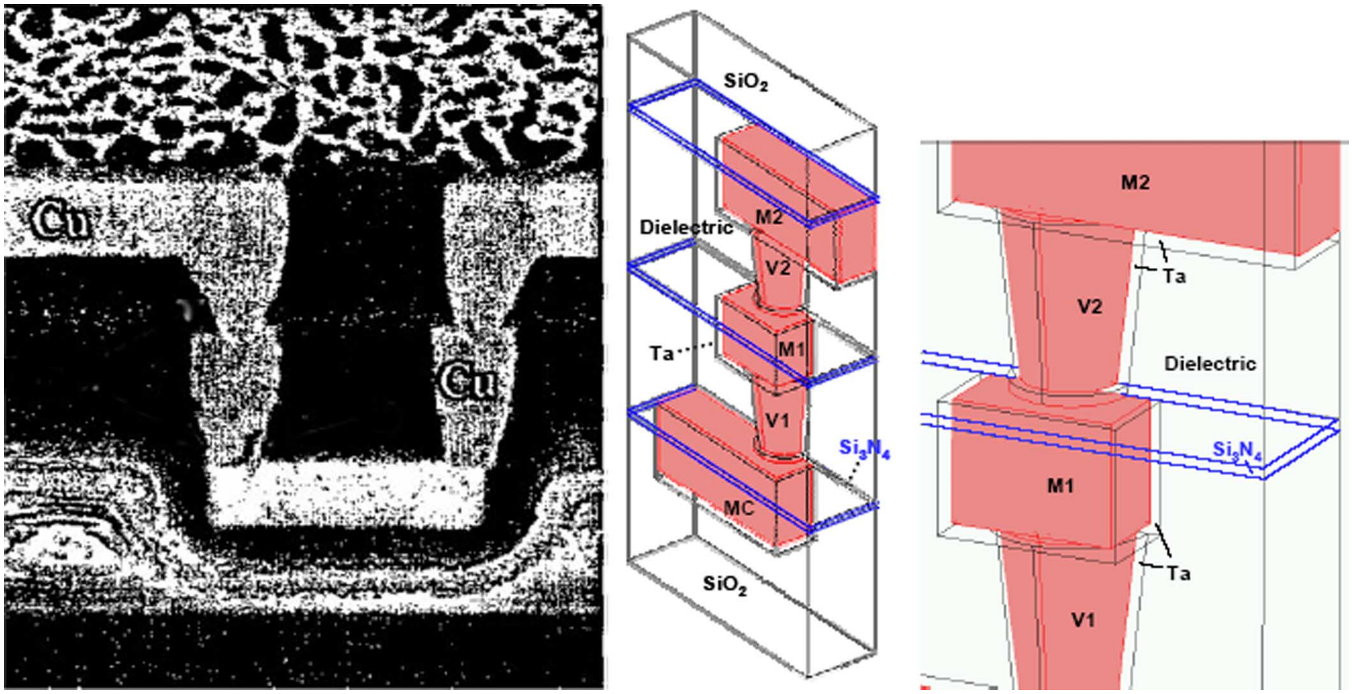


Fig. 2. Cross-sectional view of line structure, based on [1]. Test structure contains four sets of lines, with width/pitch (W/P) of 1.0/2.0, 0.6/1.2, 0.4/0.7, and 0.25/0.55 μm. (b) Schematic of test structure, substructures are labeled as used in text. Due to symmetry, only one-half of the Cu line in the pictured structure on left is shown in schematic. Schematic has a finite thickness with appropriate boundary conditions (see text).

assume idealized structures. Fig. 2 is a schematic cross-sectional view of the lines [2]. A 0.5-μm thermal oxide is grown on the top of 8-in wafers. A blanket dielectric, TEOS oxide, is deposited on the Si wafers. The 0.6-μm-deep trenches are patterned in the dielectric layer, and a 40-nm Ta barrier layer is deposited prior to the deposition of a Cu seed layer and Cu electrochemical deposition (ECD). The excess Cu over the lines is removed by chemical mechanical polishing (CMP). Then, a 50-nm SiN_x capping layer is deposited, followed by the deposition of a 0.4-μm-thick TEOS oxide layer. As described in [2], *in situ* XRD measurements were performed to determine stresses in the Cu during a temperature ramp (at a rate of 2 $^\circ C/\min$) from 400 °C to 25 °C. Details of the XRD measurements can be found in [2]. Strains are determined from changes in the lattice parameters [25].

In the second validation study, we examine the reliability of via chain structures fabricated using low-k dielectrics. Both Filippi *et al.* [3] and Edelstein *et al.* [5] performed thermal cycle tests on via chain structures. SiLK was used as the dielectric in [3], and carbonized glass (SiCOH) was used as the dielectric in [3]. The geometry of the via chain test structures used in [3] is shown in Fig. 3 (SEM on the left). The geometry of the test structures was not completely specified, and we chose reasonable values for some dimensions. In addition, we assume the structures used by Filippi *et al.* [3] and Edelstein *et al.* [5] have the same geometry.

As described in [3], the repeated unit in the test structure consists of three metal levels (MC, M1, and M2) connected by two level of vias (V1 and V2). MC and M2 are local interconnects in the chain, and M1 is a landing pad in the stacked via structure. V1 connects MC and M1, and V2 connects M1 and M2. In the experiment, the chain is 50 units long. Some of the following dimensions of the metal and barrier films were not reported and had to be assumed. Metal dimensions do not include liner thicknesses. Interconnects (M2 and MC) are taken to be 0.35-μm-long 0.31-μm-wide lines, with an assumed height of 0.25 μm. M1 is a 0.31 by 0.31 μm square landing pad, with an assumed 0.25-μm height. The vias are taken to be tapered cylinders 0.22 μm (assumed) in diameter at the bottom and 0.3 μm in diameter at the top, and 0.35 μm tall. V1, V2, M1, and M2 are Cu embedded in the low-k dielectrics (SiLK or SiCOH). MC



(a) (b) (c)

Fig. 3. (left) Cross-sectional SEM image (adapted from Filippi *et al.* [2]) of stacked via test structure using SiLK. (middle) Schematic of test structure, with substructures labeled as used in text. Note that only one quarter of pictured structure on left is shown in schematic in middle. (right) Localized details of test structure.

is tungsten embedded in silicon dioxide. Metal lines are passivated with a Si_3N_4 cap layer. Interconnects and vias are covered with Ta-based conductive liners at their bottoms and side walls. Barriers and capping layers are taken to be of uniform 20-nm thickness.

Both via chain test structures were cycled between -65 °C and 150 °C at rates of 22 °C per minute (up) and -14 °C per minute (down). Filippi *et al.* observe statistically significant distributions of electrical failures of the SiLK-based system, with approximately 50% of samples failing by 1000 cycles (as estimated from [3, Fig. 5]). No failure was observed in the SiCOH system after 1000 cycles [5].

III. FE Model

We constructed models in FEMLAB [26] for FE analyses of the test structures described previously, as well as for the 3-D IC study. For any such study, one needs to decide upon the material models and the parameter values to use in those models. We focus on the model structure in this paper and avoid getting side-tracked by questions that cannot be definitively answered at this time. For example, the mechanical properties of thin films can be significantly different from bulk samples of the same materials. This implies that they can depend upon the details of the system under study and can vary as a function of position in the system. We opt for simple material models over complex models, as the models and their parameters change with time. The materials are assumed to be isotropic and linear elastic and are given in Table I [4], [7], [9], [19], [27]. We use a static analysis, which assumes that the structure is in thermal equilibrium at the both the initial and final temperatures. This is a reasonable approach to start such stress analysis; however, it does not take into account transient aspects of the temperature trajectory. We also assume perfectly adhesive films and ideal material interfaces.

Linearity of deformation with temperature change has been confirmed for Cu by XRD measurements over a large temperature range (25 °C–400 °C) [19]. Other material properties for Cu and all properties for barriers and dielectrics are also considered to be temperature independent within the temperature ranges considered. We use material properties at room temperature. The main reason is that temperature dependencies are not clear and/or their inclusions affect the computed results well within what we consider to be the uncertainties of the model. For example, [28] reported a constant CTE of 63.6 ppm/°C over the temperature range from 25 °C to 180 °C. However, the manufacturer of BCB, Dow Chemical, shows a temperature-dependent formula for CTE: $\mathrm{CTE}(T) = 39.4 + 0.127T$, where T(°C) is temperature over a range from 40 °C to 175 °C [29]. In [28], the Young's modulus of BCB was found to decrease from 2.5 to 0.3 GPa from 25 °C to 180 °C. This value is lower than 2.9 GPa [27] reported by Dow (assuming room temperature).

In order to evaluate if computed stresses indicate possible reliability concerns, a critical value for potential mechanical and electrical failure must be chosen. Interconnect structures must be long-term stable under continued thermal cycling, so we have chosen a most conservative estimate for a failure criterion, the onset of plastic yield. There is a large variation of yield strength of Cu in the literature, and it has been reported to depend upon film thickness, grain size, and temperature. For example, the

TABLE I
MATERIALS PROPERTIES IN STUDIES

Material	E, GPa	υ	CTE, ppm/°C
Cu	115	0.343 [1]	17.7 [1]
Ta	185.7 [1]	0.342 [1]	6.5 [1]
SO_2	71.4 [1]	0.16 [1]	0.51 [1]
SiN_x	220.8 [1]	0.27 [1]	3.2 [1]
Si	131 [1]	0.278 [1]	2.61 [1]
SiLK	2.5 [4]	0.4 [4]	66 [4]
W	344.7 [7]	0.28 [7]	4.5 [7]
MLM	84.5 [a]	0.215 [a]	5.67 [a]
SiCOH	16.2 [9]	0.19 [b]	12 [26]
BCB	2.9 [27]	0.34 [27]	52 [27]

[a] Computed values based on 30% vol. Cu and 70% vol. SiO_2
[b] Assuming Poisson's ratio is same as that of fluorosilicate glass (FSG)

yield strength of thin film Cu was measured from 225 MPa for 3.015-μm film thickness to 300 MPa for 0.885 μm thick using a bulge test [30]. In another study, micromachined Cu thin film beams were deflected until inelastic deformation was detected [31]. The yield strength was estimated to be 2.8–3.09 GPa.

In order to proceed in the presence of this diversity, we consider a range of yield strength. The yield strength of bulk materials is the lower limit and one estimate for thin films is the upper limit. For bulk Cu, the yield strength of hard drawn Cu wires was measured between 414 and 483 MPa [32], assuming the yield strength is the same as the tensile strength. In the following, we use 500 MPa, a round number, to be the lower limit of yield strength at room temperature for very small Cu films with small grains. For the upper limit of yield strength, we assume the grain size is same as the via diameter in our planar IC MLM study [3], i.e., 0.22 μm. Using a Hall–Petch formula [30], the upper limit of yield strength for Cu interconnects at room temperature is about 600 MPa. For conservative estimates, the lower limit plays a role when comparing computed stresses with experimental results [3], [5].

The presence of the relatively weak BCB layer in the system is also a potential reliability concern. Critical tensile stress in BCB has been sited by the manufacturer as 85 MPa, significantly less than our critical value for copper. However, because BCB is a polymeric material with considerable toughness (reported critical elongation percentages of 6%–8%) and because it does not have to act as a conducting pathway, some plastic deformation is allowable without failure of the circuit. Further, the BCB layer in bonded wafers with interwafer vias has been shown to be stable, without exhibiting mechanical failure of the BCB layer under processing [33].

The total strain tensor ϵ_{ij}, i.e., the jth component of the gradient of displacement in the ith direction, in systems that are under mechanical stress and that undergo a temperature change can be decomposed into elastic and thermal components [34]

$$\varepsilon_{ij} = \varepsilon_{ij}^{e} + \varepsilon_{ij}^{T}. \tag{1}$$

The elastic strain component can be expressed as [34]

$$\varepsilon_{ij}^{e} = \frac{1+v}{E}\sigma_{ij} - \frac{v}{E}\sigma_{kk}\delta_{y} \tag{2}$$

where E and ν are Young's modulus and Poisson's ratio, respectively, σ_{ij} is the stress tensor, and δ_{ij} is the Kronecker delta. The thermal strain component is [34]

$$\varepsilon_{ij}^{T} = \alpha \Delta T \delta_{ij} \tag{3}$$

where α is the coefficient of thermal expansion and ΔT is the change in temperature of the solid from a stress-free reference temperature. Substituting (2) and (3) into (1), we obtain a constitutive equation for isotropic linear elastic materials [34]

$$\varepsilon_{ij} = \frac{1+v}{E}\sigma_{ij} - \frac{v}{E}\sigma_{kk}\delta_{ij} + \alpha \Delta T \delta_{ij}. \tag{4}$$

We use linear tetrahedral finite elements within FEMLAB to obtain numerical solutions to (4). We locally refine the mesh in appropriate regions to resolve the stresses and strains in certain substructures, such as the barrier and capping layers, as shown in Fig. 3. Mesh refinement adequacy is confirmed by the lack of significant change in local stresses upon varying the mesh density used to solve the model.

The stresses and strains calculated with this model are only valid when plastic deformation has not yet occurred, and thus computed values of the stresses above the yield strength of Cu are not meant to imply that these stresses actually occur in the system. Rather, the presence of such stresses in the computed solution implies that plastic deformation will occur and that reliability is a concern in such systems.

IV. VALIDATION STUDIES

A. Cu Line Study

For the XRD-based study of Cu lines, we use a half line model as shown in Fig. 2 due to multiple parallel lines in the test structure and their periodic nature. A similar model was used in [2] to compare against XRD data. In this half-line model, the left and front faces of the model are only allowed to shift within their planes due to symmetry. The right and back plane experience thermal contraction along the normal direction of each plane by an amount given by the expression $\Delta L = \alpha L \Delta T$, where L is the total width or depth of the model; ΔT is the temperature change; and α is coefficient of thermal expansion of Si substrate. The bottom of the model is fixed vertically but

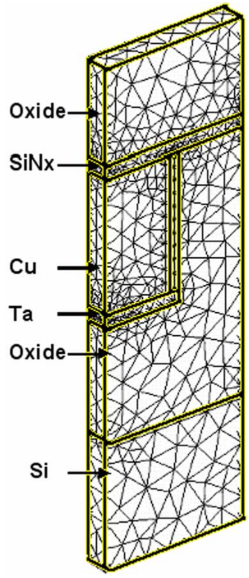


Fig. 4. Tetrahedral FE mesh with local refinement used to solve (4) on test structure, while resolving stresses in barrier and capping layers. Heavy lines indicate materials interfaces.

can displace laterally. The top surface is free to move. The FE model in this paper is shown in Fig. 4. We choose the thickness of the Si substrate to be 0.5 μm and the depth to be 0.1 μm. The calculated stresses do not depend on these values when using the above-mentioned boundary conditions.

We present computed results of Cu line stresses as stress slopes $d\sigma/dT$, rather than stress values, as in [2]; i.e., that his how the XRD-based data are presented. This removes concerns about our lack of knowledge of the residual stresses at the initial elevated temperature (400 °C). Stress slopes emphasize thermal stress characteristics. Our simulation results agree very well with the computed stress slopes reported in [2].

As shown in Fig. 5, the stress slopes along the Cu lines and normal to the wafer surface are in reasonable agreement. However, the stress slope across the line is underpredicted by approximately 30%. In an effort to explain the lack of agreement between computed lateral stresses and those computed from measured strains, we varied several aspects in our model. It should be noted that cross sections of the lines were not provided in [1] or [20], so we do not have as much information as we would like in order to improve the structural aspects of our model. We tried several things in order to evaluate the sensitivity of the computed stress slopes to those changes, including:

1) tapering the cross section of trenches; i.e., keeping the nominal line width at the top, we decreased or increased the width of the bottom of the trench;
2) tapering the barrier layer on the side wall of the trench, i.e., decreasing the thickness of the barrier with depth;
3) varying the Young's modulus of Cu near the seams formed during ECD to higher (double) and lower (half) the Young's modulus of Cu;
4) variying the mechanical properties of the Ta barrier (i.e., doubling/halving Young's modulus of Ta);

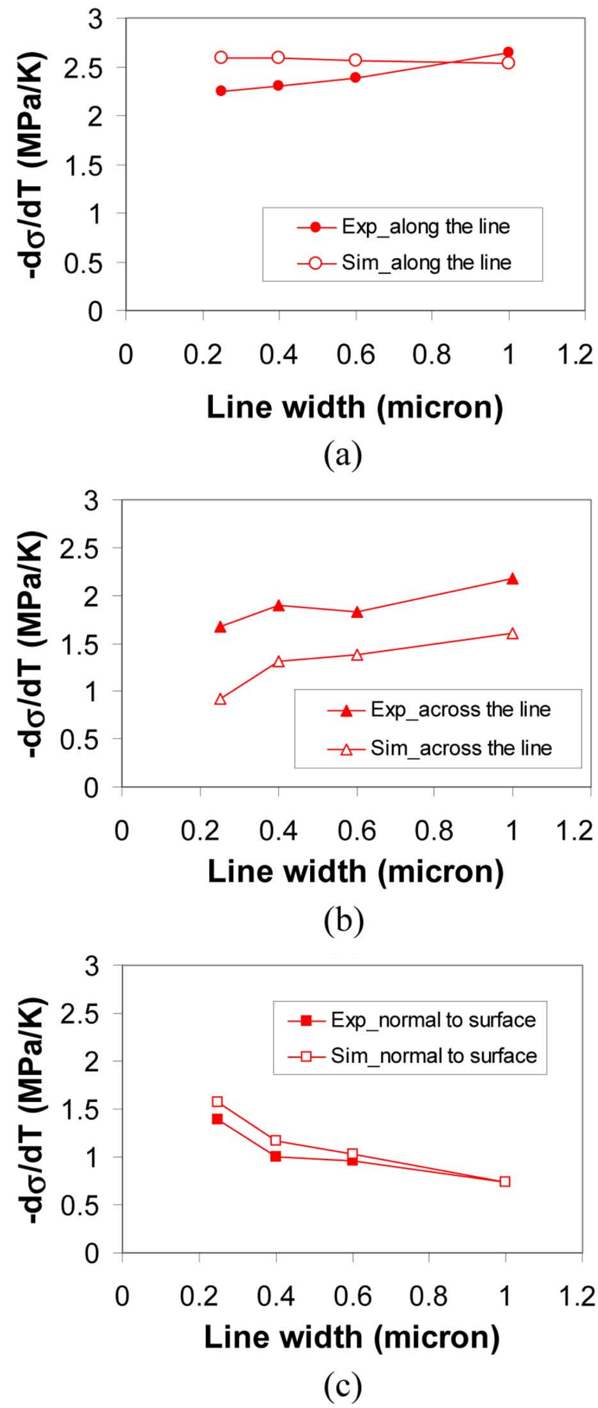


Fig. 5. Comparison between XRD derived and computed stress slopes.

5) using various combinations of anisotropic Young's moduli for $\langle 111\rangle$ Si and $\langle 100\rangle$ Si ($E\langle 111\rangle = 191$ GPa, $E\langle 100\rangle = 67$ GPa).

These changes, among others, did not significantly improve the agreement between computed and experimental stress slopes.

On the other hand, when we use Cu lines that are 0.1 μm wider than reported, then the stresses across the line direction are closer to the data ($\sim$20% difference) while the agreement in the other two directions does not change substantially. This variation seems like a large error in line size, and without SEM

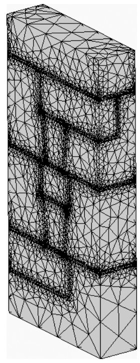


Fig. 6. Tetrahedral FE mesh with local refinement used to solve (4) on test structure, while resolving stresses in barrier and capping layers.

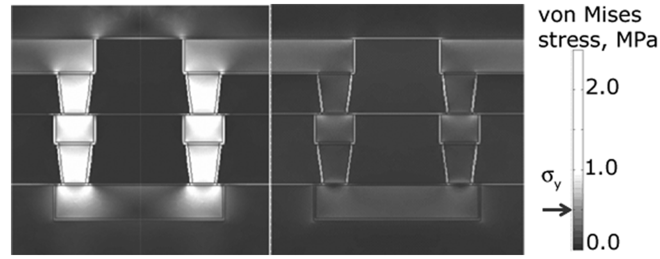


Fig. 7. Computed von Mises stresses in SiLK-based system (left) and SiCOH-based system (right). Note that almost of all of V1, M1, and M2 Cu is above top range of yield strength (600 MPa) in SiLK system. None of the Cu is above nominal yield strength (500 MPa, marked with arrow on scale) for SiCOH system.

cross-sectional information we can only say that the results are sensitive to any errors in linewidth.

The model modifications summarized are not able to completely explain the disparity between computed stresses and those computed from XRD measurements. As with any experimental studies, there are also uncertainties in the XRD based determination of the stress slopes [25], [35]. We could speculate further regarding both the detailed structure of the samples, the parameter values used in the model, and/or experimental uncertainties; however, we are basically satisfied with the ability of the model to compute stresses in metal lines.

B. Via Chain Study

For the via chain structures, we compute stresses due to a single temperature change in the same range as that performed in the thermal cycle experiments [3], [5]. Our geometric model represents one quarter of the via-chain repeat unit as shown in Fig. 6. The periodic nature of the repeating via-chain unit is represented by applying symmetric boundary conditions on the sides of the simulation cells that cut transversely through the chain direction (the "transverse sides"), allowing points on these sides to remain in-plane. The simulation cell sides that cut vertically through the chain and are parallel to its direction (the "parallel sides") are also treated symmetrically. This set of symmetries means that the full linewidth of the interconnects and circular cross section of the vias are accounted for and that the model represents a large bank of via chains fabricated in parallel lines on a 0.6-μm pitch. The boundary conditions are same as described in the section of Cu line studies. The top surface is free to move, and the structure is taken to be at a stress-free reference state at its maximum temperature, 150 °C.

Fig. 7 shows calculated von Mises stresses throughout both the SiLK-based and SiCOH-based structures, and Fig. 8 (right) shows the von Mises stresses in just the Cu on an expanded scale for the SiLK system. Fig. 8 (left) shows a failed V2 via, after the dielectric was removed by oxygen ashing [3]. The crack appears to coincide with a shear plane [3]. Points where the von Mises stress exceeds the yield stress of Cu indicate sites of potential failure. The thermo-elastic model predicts that the stresses in the Cu vias in SiLK exceed the upper limit of yield strength of Cu (600 MPa). However, the stresses in Cu in SiCOH are below the lower limit of yield strength (500 MPa). The CTE of SiLK over temperature ranges common to BEOL processing is about 66 ppm/°C [36], which is considerably larger than that of Cu (approximately 17 ppm/°C [22]) in the same temperature range. It is not surprising that the strain induced in Cu metallization during temperature changes can lead to degradation and failure. Compared with SiLK, the CTE of SiCOH (12 ppm/°C [9]) is close to that of Cu, and the thermal stress in the Cu via is substantially reduced.

Fig. 9 shows some results of parametric studies that we performed to determine the sensitivity of our computed results to the model parameter values used; i.e., we vary materials' properties within reasonable ranges to identify the effects on thermal stresses. Computed von Mises stresses sampled at 0.115 μm above the M1 along the axial line of top via (point P in Fig. 8), which is or is close to the maximum stress in the top via, increase with increasing Young's modulus, CTE, and Poisson's ratio (Fig. 9). This figure, in combination with yield strength, provides a failure criterion for Cu vias under different combinations of materials properties. As is reasonable, the figure also suggests that dielectric materials with lower Young's modulus, CTE, and Poisson's ratio can reduce the stress in the Cu interconnect. Fig. 9 reveals the effects of the properties of the dielectrics on the stresses and can be used to evaluate proposed dielectrics.

One of the concerns in this simulation study is the treatment of the very thin barrier and capping films; e.g., we assume they are of uniform thickness and ignore any granularity. That is consistent with using idealized geometries for the vias and lines. However, small variations will have a larger effect on them than for the thicker materials. It is also not clear how the various material interfaces impact results. In order to evaluate the impact of the assumptions, we gauged the effect of including barrier and capping films on the stresses calculated in our simulations. As long as the thickness of barrier layers is thin, the inclusion of thin barrier layers does not substantially change the stress distributions in the vias. For example, our simulations show that the stresses along the center of the top and bottom vias are about 10% higher when the volume occupied by barriers is replaced by Cu (Fig. 10). It is higher because barrier films are stiffer than Cu, and the stresses are higher in the barrier films when they are included. It should also be noted that the computed stresses in the barriers surrounding the vias are larger than the yield strength of Ta ($\sim$ 350 MPa) [37]. In that case, the Ta barriers will fail, perhaps leading to the diffusion of Cu into surrounding dielectrics. This barrier failure mechanism is out of the scope of

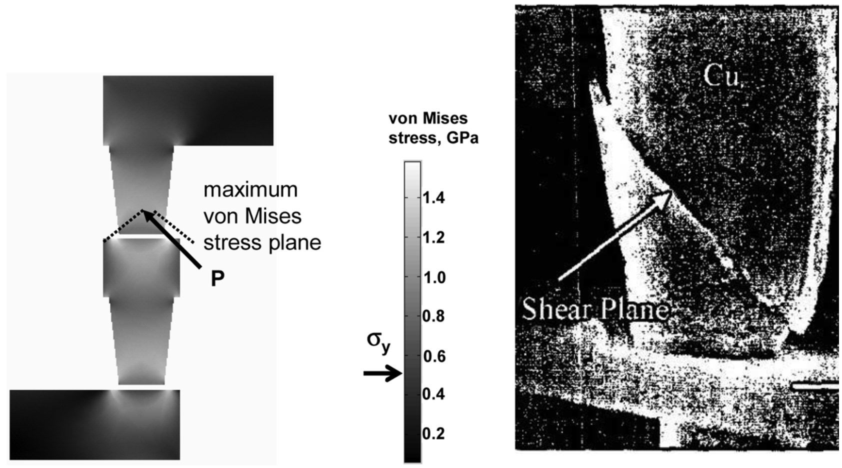


Fig. 8. (left) Computed von Mises stresses in Cu sections of test structure using SiLK. Note stress concentrations near top of M1 landing pad and bottom of V2 via. Point P is bottom center of V2 via, where stress tends to concentrate due to small cross section and denotes location where stress is calculated in Fig. 7. Arrow in scale indicates nominal yield strength of Cu. (right) SEM image from Filippi *et al.* showing shearing and cracking along bottom of V2 via.

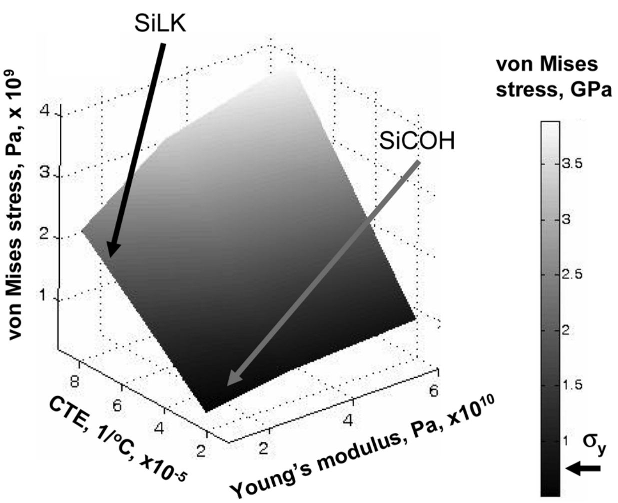


Fig. 9. Calculated stress as function of Young's modulus and CTE at point P shown in Fig. 8. Poisson's ratio is 0.4. Arrow on scale indicates nominal yield strength of copper.

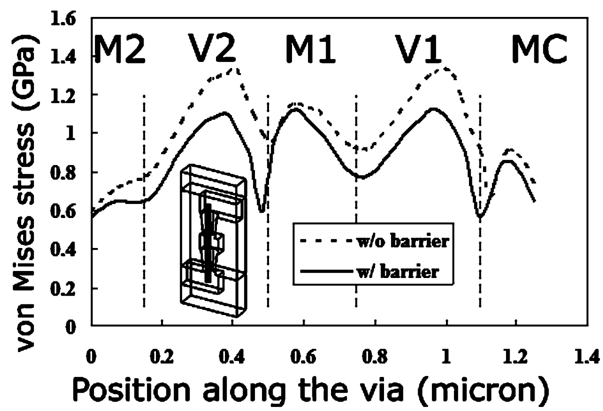


Fig. 10. Stress distributions along via embedded in SiLK with and without barrier layers. Thick dark line in insert shows positions along via where stresses are plotted.

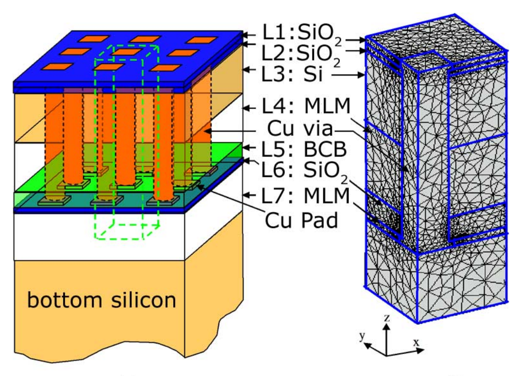


Fig. 11. 3-D structure. (a) Schematic of several cylindrical vias connecting square landing pads through layered structure (not to scale). Dashed block is unit cell. (b) Typical FE mesh. Taking advantage of symmetry, only one quarter of a via and pad has been included in geometric model.

this paper. From the mechanical failure point of view, the focus of this work, the barrier layers cannot sustain large force due to their small thicknesses. In that case, ignoring them may provide a good model for the local mechanical structure. The barrier and etch stop layers are not considered in the models of 3-D structures discussed in Section V.

V. Thermal Stress in 3-D Structures

We perform simulations to calculate stress in a representative unit cell. The model consists of seven layers with embedded Cu interconnects (Fig. 11). In order to derive design-related parameters and make the simulation tractable, we choose a repeated unit cell in the 3-D IC structures (Fig. 11). The top Si wafer (L3) is thinned to 10 μm thick and bonded to the bottom wafer using a 2.6-μm-thick BCB (L5). The bottom Si wafer is not included in the unit cell model to keep the number of elements down. We checked the computed stress using the same boundary conditions as used in the XRD-based validation study discussed previously and found only a small difference (less than 5%) of stresses in the Cu vias. Circular Cu vias, with an assumed 23.6-μm height, connect 10-μm-thick MLMs (L4, L7) in the two wafers. Square landing pads are used at the end of Cu vias

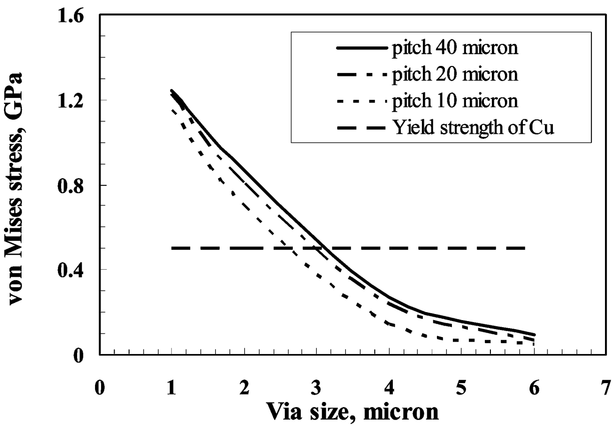


Fig. 12. Computed von Mises stress in Cu vias of different size and different pitches for BCB thickness of 2.6 μm.

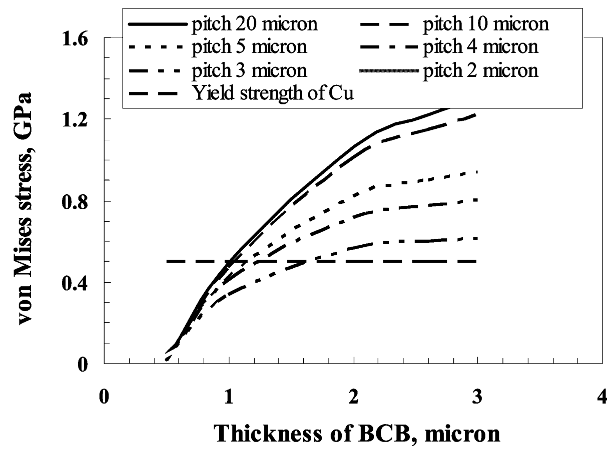


Fig. 13. Computed von Mises stress in 1-μm Cu vias with different BCB thicknesses and pitch sizes.

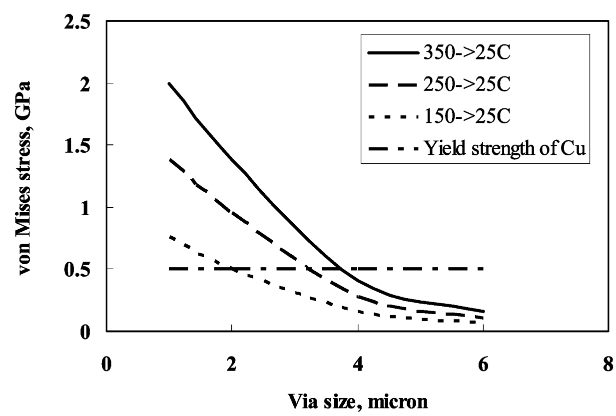


Fig. 14. Maximum von Mises stress in Cu vias of different size cooling from 150 °C, 250 °C, and 350 °C to 25 °C.

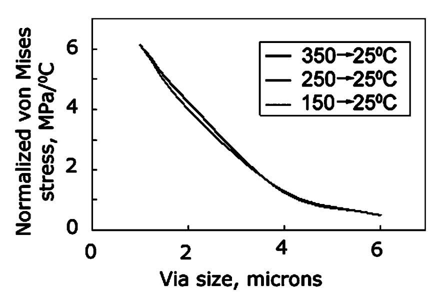
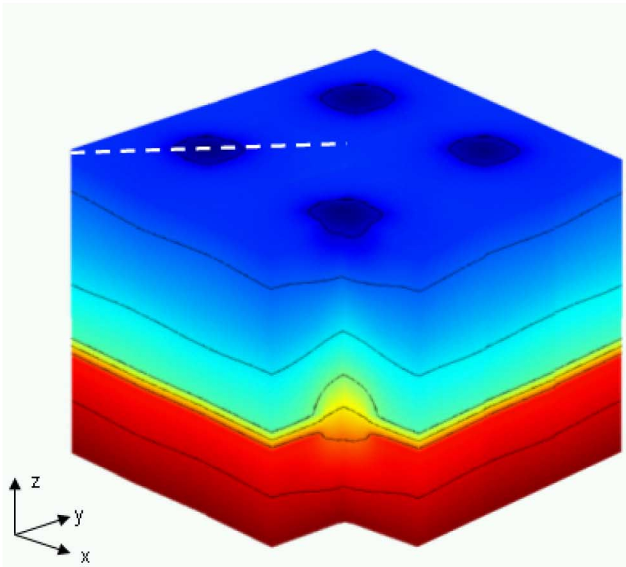


Fig. 15. Normalized maximum von Mises stress by temperature change in copper vias of different size, cooling from 150 °C, 250 °C, and 350 °C to 25 °C.

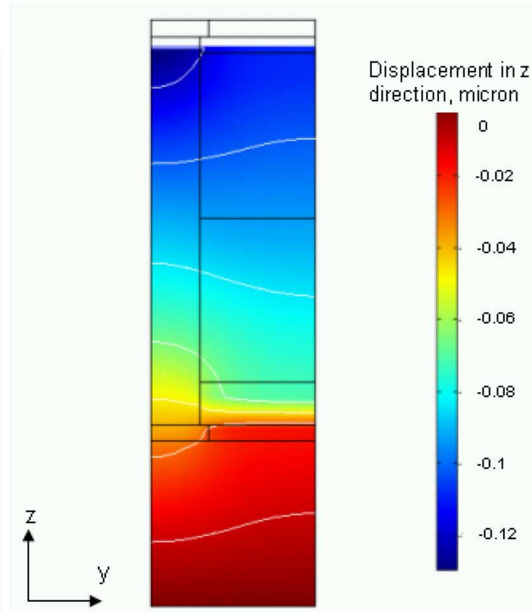
and embedded in the oxide layers (L1, L2, L7). Due to symmetry in the x-z and y-z planes (Fig. 11), only one quarter of a unit cell is used in the simulations. Barrier materials are not considered in the model due to their small thicknesses, and as discussed, their presence does not significantly affect computed stresses in our model of planar IC structures. The structure is assumed stress-free at 250 °C (wafer bonding temperature). The material properties and dimensions of the model are given in Table I. Young's modulus of BCB is found to decrease with temperature from 2.5 GPa at 25 °C to 0.3 GPa at 180 °C [28]. To be conservative, we use the value at 25 °C, which will tend to overpredict the stresses in the Cu. We use the room temperature value of CTE for BCB, which wafer-curvature experiments [38] show to be largely temperature independent. For other material properties of pure materials, e.g., Poisson's ratio of Cu, we use material properties widely reported at room temperature. The material properties of the MLM layer are determined by the volume average of each component (30% Cu and 70% SiO_2) [39]. For mechanical boundary conditions, the side walls of the unit cell are only allowed to shift within their respective planes due to symmetry, as is the bottom surface, which is assumed perfectly bonded to the thick Si substrate below. The top surface is free to move. As seen in the right half of Fig. 11, the FE mesh is refined in areas where large stress gradients are expected.

Fig. 12 shows the maximum von Mises stress in Cu vias between 1 and 6 μm in diameter at pitches of 10, 20, and 40 μm, for an initial temperature of 250 °C and a final temperature of 25 °C. The order of CTEs of the materials in the stack materials is: BCB $>$ Cu $>$ MLM $>$ Si $>$ SiO_2, and the BCB layer is quite thin. A back-of-the-envelope calculation shows that the unconstrained thermal contraction of the Cu vias were they not in the stack ($\sim$ 84 nm), would be significantly greater than the unconstrained thermal contraction of a stack without the Cu vias ($\sim$ 55 nm). Thus, in the 3-D stack in which the vias are coupled to the other materials, the Cu vias act to constrain the vertical expansion of the entire structure. At each pitch, the maximum von Mises stress in the Cu increases as the via size decreases. This trend makes sense because the cross section of the Cu decreases while the amount of stack it must constrain increases. The calculated von Mises stress decreases with decreasing pitch at constant via size, as the force per unit area is distributed to more vias. Our simulations indicate that plastic deformation of the vias is a concern. Combining yield strengths of Cu discussed previously, we can estimate some target values for design parameters. The lower limit of yield strength (500 MPa) can be used for a conservative design. As shown in Fig. 12, for a BCB thickness of 2.5 μm, the via size should be larger than 3 μm at a pitch of 10 μm, and 3.5 μm at a pitch of 40 μm.

Simulations were performed to evaluate the use of BCB bonding with small Cu vias at various pitches, such as might be used to bond logic and memory circuits. We assume a via diameter of 1 μm, vary the via pitch from 2 to 20 μm, and vary the BCB thickness from 0.5 to 3 μm. Fig. 13 shows the computed



(a)



(b)

Fig. 16. (a) Contour and isosurface (black curves) of vertical displacement (z) calculated for 6 μm vias, a pitch of 20 μm, and a BCB thickness of 2.6 μm. Structure contains 15 unit cells shown as in Fig. 11(b). Unit cell is removed (front corner) to reveal internal displacements. Dashed white line along diagonal direction on top surface is used in Fig. 17. (b) Side view of vertical displacement. White curves are isosurface of vertical displacements. Black lines delineate undeformed structure.

stress at 0.5 μm along the center line above the lower landing pad and indicates that the thickness of BCB should be less than 1 μm, using yield strength of Cu 500 MPa, to avoid plastic yield of Cu vias for pitches larger than 3 μm. This is due to the relatively high CTE of BCB compared to Cu. Although the "unconstrained stack" discussed previously contracts approximately 55 nm, approximately 60% of that motion comes from the 2.6 μm of BCB. Over that same length, Cu contracts only 10 nm when unconstrained, so when they are coupled to each other, a significant compressive stress is induced in the lowest part of the via during cooling. By reducing the thickness of the BCB layer, this expansion can be reduced in magnitude, and the stress in the bottom of the via reduced. For 1-μm diameter via and 2-μm pitch, computations indicate that there is no concern about plastic yield of Cu vias for the BCB thicknesses studied. As the pitch decreases, the volume of BCB around the vias is also reduced. As in the earlier discussion on the effect of BCB thickness on the stresses, smaller pitches (higher via densities) correspond to less BCB volume and smaller stresses in the vias.

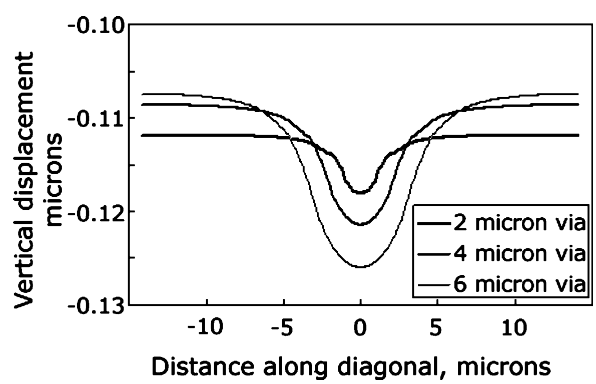


Fig. 17. Vertical displacement along a line in diagonal direction for 20-μm pitch (see Fig. 16). Note expanded vertical axis.

We also studied the effect of imposed temperature changes on the thermal stresses. Simulations of different initial stress-free temperatures, 150 °C and 350 °C and cooling down to room temperature were performed to study thermal stress levels. The maximum von Mises stress in the Cu via is shown in Fig. 14. As expected, the stress increases as the initial temperature increases. In [3], a threshold temperature range ΔT_c of 145.3 °C [3, Table I] was found from statistical analysis of mean-cycle-to-failure, N_{50}, as a function of temperature changes ΔT ($T_{\max}$ was 150 °C, ΔT varied from 150 °C to 300 °C) [3, Fig. 8]. Below this threshold temperature, the test structures did not fail. Our simulations also show that the magnitude of temperature change is a major factor determining the stress level in the interwafer vias. If the structures cool down to room temperature, the maximum processing temperatures determine the thermal stress levels. The stress data are normalized by the temperature difference between initial and final temperatures and plotted in Fig. 15. As expected for a linear analysis, the stresses at different initial temperatures collapse into a single curve. Therefore, when thermally induced stress information is available for some temperature ranges, it is possible to extend this information into other temperature windows through this procedure, assuming the model parameter values are reasonable for the temperature range of interest.

Issues in addition to stresses in Cu can be of interest. As discussed above, Cu vias contract more than the surrounding materials, thus pulling down on these neighboring materials to which

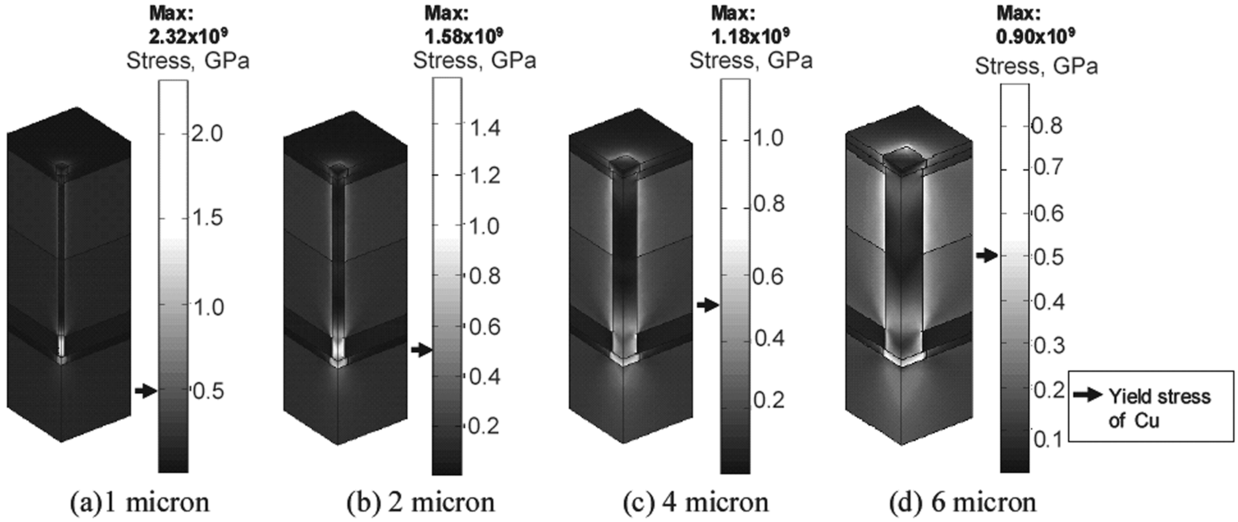


Fig. 18. Computed von Mises stresses in 3-D IC structure with via size of 1, 2, 4, and 6 μm for 20 μm pitch. Arrows on scales indicate nominal yield strength of Cu.

they are coupled. For example, Fig. 16 shows that the top surface of a 6-μm-diameter via is lower than the surrounding material. This roughness, which can be on the order of 100 nm, may affect the downstream processes, and the adhesion of layers on deposited after bonding and thinning. Fig. 16 also shows the side view of displacement in the vertical direction. The same simulation was repeated with different via sizes, ranging from 2 to 6 μm. The resulting vertical displacements of the top surface are shown in Fig. 17. The size of the depressed area increases with via size due to both the larger via diameters and the increased constraining forces transmitted by the via.

At constant via pitch (we use 20 μm for illustration), the maximum von Mises stress shifts from the Cu vias to the top Si substrate as the via size increases as shown in Fig. 18. At relatively low via densities ($< 6\%$ area fraction), the maximum stress was observed in the Cu vias. Results suggest that Cu vias smaller than a critical size will yield plastically. As the via density increases ($> 10\%$), the maximum stress gradually shifts to the top, thinned, Si substrate. Potential yield or failure locations (e.g., failure in interconnect versus Si substrate) are also related to this critical via density. For brittle materials, such as Si, the critical fracture stress is a function of any flaw or existing damage to the specimen [40]. Such damage can be caused by wafer processing, such as grinding or etching. A carefully polished Si specimen has an average fracture stress as high as 800 MPa [41]. On the other hand, a backside processed wafer has the fracture stress only 175 MPa [41]. In our simulations, the maximum stresses in Si occur in the vicinity of the interface between the Si and the Cu vias. As the via density reaches approximately 10%, the stress in Si exceeds this estimate of the critical fracture stress. The shift in the location of the maximum stress can be understood by considering that the Cu via is in parallel with other layer materials, i.e., the Cu vias need to deal with the forces exerted by their surroundings. As via size increases or via density increases, the total Cu area increases. So, stresses are lower for the same forces.

Several areas can be identified for improving our analyses. First, in the via chain study, we partially validated our model using planar IC data [3] by comparison with experimental failure information—a qualitative response. The stress level at which vias fractured rather than just underwent plastic deformation was not studied in [3], and we are not aware of any such data in the literature. Thermal stress measurements using XRD of Cu lines passivated with TEOS oxide and methyl silsesquioxane dielectrics show that, depending on the processing conditions, initial stress at 400 °C may be on the order of 100 MPa [2]. A careful evaluation of initial stresses is necessary for proper prediction of thermal stresses. In addition to helping to evaluate assumptions in our model, this is an important part of understanding the mechanical reliability of 3-D structures. Second, in our simulations, materials are assumed to be homogeneous and no information on microstructure is explicitly included. It has been shown that stress levels increase with increasing average grain size for 0.35 μm Cu lines with passivation [42]. Grain boundaries are interfaces which serve as stress relaxation sites. Increasing the grain size reduces the number of grain boundaries and leads to the higher stresses. A model linking stress and microstructure would be useful to our analysis. In addition, any gaps in the structure, such as might occur during deposition in features, will decrease thermally induced stresses. In order to improve our understanding of topography, microstructure, and/or failures of Cu vias during thermal cycles, process models, e.g., EVOLVE [43], [44] or PLENTE [45], [46] should be incorporated into the computations. Third, the role of barrier layers on the mechanical strength of vias needs to be addressed further. Finally, the static analysis performed in this paper does not account for any effects of creep or fatigue. Von Mises stresses are used to predict plastic deformation, but not all structures destined to fail due to thermally induced stress do so with a simple temperature change, nor does plastic deformation always lead to electrical failure.

VI. Conclusion

We presented a FE-based model in order to evaluate whether thermally induced stresses in Cu interwafer vias used in

BCB-bonded (wafer level) 3-D ICs are a reliability concern. Our model was first tested against two types of experimental data; one was XRD determined stresses in Cu lines, and the other was failure analyses of via chain structures with Cu interconnects and low-k dielectrics. Computed von Mises stresses, assuming a single cooling step of the same magnitude as the experiments, are at a maximum near the bottom of vias in the planar-IC, via-chain test structure, which coincides with experimental observations of failure when SiLK was used. No failure in the vias is predicted when SiCOH was used as the dielectric, which is consistent with experiments.

We then calculated thermal stresses in 3-D IC structures bonded with BCB using a process developed at RPI, due to changing temperature from 250 °C to 25 °C. Simulations show that the von Mises stresses in the Cu vias decrease with decreasing pitch at constant via size, increase with decreasing via size at constant pitch, and decrease with decreasing BCB thickness. We found that further study is indeed warrantedm i.e., the stresses in Cu vias either exceed or are close to the yield stress for temperature changes similar to those used in BEOL processing and interwafer via pitches and via sizes that may reasonably be used in 3-D ICs. Our results also support the goal of using thin BCB. In general, thin BCB has been targeted in order to keep via aspect ratios down; however, it helps also keep the via stresses due to BCB contractions and expansions down as shown by Fig. 9 and discussed in Section IV. Target values for design parameters, e.g., interwafer via size, pitch, and the thickness of BCB, were estimated. An additional concern is highlighted by the current work as well; thinned Si in these stacked wafers may fail.

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Jing Zhang received the B.S. degree (honors) from the University of Science and Technology, Beijing, China, in 1996, the M.S. degree (honors) from Beijing University of Aeronautics and Astronautics, China, in 1999, and the Ph.D. degree from Drexel University, Philadelphia, PA, in 2004.

During 2004 and 2005, he was a Postdoctoral Research Fellow at the Focus Center, Rensselaer Polytechnic Institute, Troy, NY. In 2005, he joined the Department of Mechanical Engineering at the University of Alaska, Fairbanks. His research interests include multiscale simulations, couple phenomena, and applications in materials processing.

Dr. Zhang was the recipient of the 2003 CPMT/Axel Madsen Scholarship from the Center for Powder Metallurgy Technology and the 2004 Best Student Poster Award from The Materials Society (TMS).

Max O. Bloomfield received the B.S. degree in chemistry from the State University of New York, Albany. He is working toward the Ph.D. degree in the Department of Chemical and Biological Engineering, Rensselaer Polytechnic Institute, Troy, NY.

He has worked for the Focus Center, Rensselaer Polytechnic Institute, and Sun Microsystems, Inc. His doctoral work focuses on developing novel computational techniques for tracking interfaces in many-phase systems, which he has incorporated into a parallel software framework, called PLENTE. These techniques have direct application to simulations of microstructure, radiological dosing, and multiphase fluid flow.

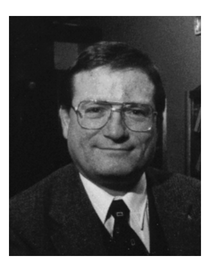
Jian-Qiang Lu (M'97–SM'01) received the Dr.rer. nat. (Ph.D.) degree from the Technical University of Munich, Munich, Germany, in December 1995.

He is an Associate Professor at Rensselaer Polytechnic Institute, Troy, NY, where he has been leading the three-dimensional (3-D) hyper-integration technology research programs since 1999. Prior to 1999, he held research and faculty positions at universities in China, Germany, and the United States. He has broad research experiences from micro-nano-electronics theory and design to materials, processing, devices, integration and packaging. His current research interests include 3-D hyper-integration of materials and systems, micro-nano-bio interfaces for future chips, electro-photonic technologies for new energy, biotech, and MEMS. He has authored/co-authored more than 150 publications and has given a number of invited presentations.

Dr. Lu served as workshop and session Chair for conferences and as a Reviewer for a number of journals. He is also a member of the American Physical Society (APS), Materials Research Society (MRS), Electrochemical Society (ECS), International Microelectronics and Packaging Society (IMAPS), and a member of the IMAPS National Technical Committee (Chair for 3-D Packaging). He was a Siemens Scholar, 1992–1996, and a scholar of the German Academic Exchange Program (DAAD), 1990–1992. He has received many awards, including a recent "School of Engineering Research Excellence Award," Rensselaer, 2005.

Ronald J. Gutmann (S'58–M'63–SM'76–F'94) received the B.E.E. degree from Rensselaer Polytechnic Institute (RPI), Troy, NY, in 1962, the M.E.E. degree from New York University, New York, in 1964, and the Ph.D. degree in electrophysics from RPI, in 1970.

From 1962 to 1966, he was a Member of the Technical Staff at AT&T Bell Laboratories, where he worked on the development of microwave components and systems for radar applications. From 1966 to 1970, he was at Lockheed Electronics Company, where he worked on beam-steering and beam-forming techniques for phased arrays, as well as at RPI and at the Rensselaer Research Corporation, developing novel semiconductor control devices. From 1970 to 2005, he was on the faculty at RPI, a Professor in the Electrical, Computer, and Systems Engineering Department, with teaching and research activities in the areas of semiconductor devices, microwave monolithic integrated circuits (MMICs), and thin-film high-density interconnection technology. He has authored or coauthored over 300 technical papers in these and related fields, presented approximately 30 invited talks at major international conferences, and has coauthored three books on IC interconnect technology: *Chemical-Mechanical Planarization of Microelectronics Materials* (Wiley, 1997), *Copper-Fundamentals for Microelectronic Applications* (Wiley, 2000), and *Chemical-Mechanical Polishing of Low Dielectric Constant Polymers and Organosilicate Glasses* (Kluwer, 2002). His recent research activities have been in three-dimensional (3-D) IC technology platforms and design, copper interconnect technology, silicon carbide (SiC) and gallium nitride (GaN) power devices, and microwave photoconductivity decay (M-PCD) characterization of recombination processes in semiconductor materials and device structures. He continues his research in 3-D IC technology platforms and design as Professor Emeritus, with a focus on redistribution layer bonding for wafer-level packaging and monolithic point-of-load (PoL) DC-DC converters for cell-based power delivery. He is a consultant and expert witness for the semiconductor, IC, and packaging industries.



Timothy S. Cale received the Ph.D. degree in chemical engineering from the University of Houston, Houston, TX, in 1980.

He is a Professor of chemical and biological engineering at Rensselaer Polytechnic Institute, Troy, NY, and a PI in the Focus Center, Rensselaer. The center was established to develop materials and processing solutions for integrated circuit design and fabrication for ten years in the future and beyond. His research focuses on transport and reaction fundamentals of processes used to fabricate integrated circuits. He developed a widely used model for transport and reaction during low-pressure deposition and etch processes. This model has been used in several computer programs to understand how process conditions affect thin film processing at the feature (micron scale). He has been interested in multiscale process modeling and simulation for a number of years. He has proposed processes in which process conditions are varied in a preprogrammed, model driven manner to optimize process or film properties. He currently focuses on simulations designed to understand microstructure formation and evolution, with a goal to predict product relevant performance of thin films.